

Transportation – Predictive Analytics for Vehicle Maintenance in Fleet Management

Industry: Transportation

Problem

Unexpected vehicle breakdowns causing delays and increased maintenance costs.

Business Value

- Reduce vehicle downtime and improve fleet availability.
- Optimize maintenance schedules to lower operational costs.
- Enhance overall fleet reliability and service quality.

Solution Overview

- **Data Analysis:** Use Excel to clean and summarize vehicle sensor and maintenance data.
- **Data Extraction:** Use SQL to query fleet operational logs and sensor readings.
- **Modeling:** Develop predictive maintenance models in Python to forecast potential failures and schedule timely maintenance.
- **Visualization:** Build interactive dashboards in Tableau to monitor vehicle health and maintenance alerts.

Tools Used

- **Excel:** Data cleaning and initial exploratory analysis.
- **SQL:** Extract vehicle sensor data and maintenance records.
- **Python:** Predictive modeling for maintenance scheduling.
- **Tableau:** Visualization dashboards for fleet health monitoring and decision support.